

$$\Rightarrow \frac{2}{6} = \frac{2n+10}{2n+8}$$

So,

$$\Rightarrow 2n+10=2$$

$$\Rightarrow 2n = -8$$

$$\Rightarrow n = -4$$

Ex # 4.4

Your Success is

Our Win

① Sum the series.

① $3+8+13+\dots$ to 16 terms.

Sol

$$\therefore S_n = \frac{n}{2} \{2a + (n-1)d\}$$

here, $a=3$, $n=16$

$$d = 8-3 = 13-8 = 5$$

$$\therefore S_{16} = \frac{16}{2} \{2(3) + (16-1)5\}$$

$$\Rightarrow S_{16} = 8(6 + 75)$$

$$\Rightarrow S_{16} = 648$$

(ii) $(-3)+(-1)+1+3+5+\dots$ to 18 term

Sol

$$\therefore S_n = \frac{n}{2} \{2a + (n-1)d\}$$

here, $a=-3$, $n=18$

$$d = -1 - (-3) = -1 - (-1) = 2$$

$$\therefore S_{18} = \frac{18}{2} \{2(-3) + (18-1)2\}$$

$$\Rightarrow S_{18} = 9(-6 + 34)$$

$$\Rightarrow S_{18} = 252$$

(iii) $4.57+4.87+5.17+\dots$ to 22 term

Sol

$$\therefore S_n = \frac{n}{2} \{2a + (n-1)d\}$$

here, $a=4.57$, $n=22$ and

$$d = 4.87 - 4.57 = 5.17 - 4.87 = 0.3$$

$$\therefore S_{22} = \frac{22}{2} \{2(4.57) + (22-1)0.3\}$$

$$\Rightarrow S_{22} = 11(9.14 + 6.3)$$

$$\Rightarrow S_{22} = 11(15.44)$$

$$\Rightarrow S_{22} = 169.84$$

(31)

$$\frac{3}{\sqrt{2}} + 2\sqrt{2} + \frac{5}{\sqrt{2}} + \dots \text{ to 15 terms} \Rightarrow S_n = 7.5 (4/\sqrt{2} + 7\sqrt{2})$$

S1

$$\Rightarrow S_n = 7.5 (10\sqrt{2})$$

$$\Rightarrow S_n = 75\sqrt{2} \quad \underline{\underline{AP}}$$

$$\therefore S_n = \frac{n}{2} \{2a + (n-1)d\}$$

$$\text{here, } a = \frac{3}{\sqrt{2}}, n = n$$

$$d = \frac{2\sqrt{2} - \frac{3}{\sqrt{2}}}{1} = \frac{5 - 2\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$\therefore S_n = \frac{n}{2} \left\{ 2 \left(\frac{3}{\sqrt{2}} \right) + (n-1) \frac{\sqrt{2}}{2} \right\}$$

$$\Rightarrow S_n = \frac{n}{2} \left(\frac{6}{\sqrt{2}} + \frac{\sqrt{2}n - \sqrt{2}}{2} \right)$$

$$\Rightarrow S_n = \frac{n}{2} \left(\frac{5\sqrt{2}}{2} + \frac{\sqrt{2}n}{2} \right)$$

$$\Rightarrow S_n = \frac{n}{4} (5\sqrt{2} + \sqrt{2}n)$$

$$\Rightarrow S_n = \frac{n\sqrt{2}}{4} (5+n) \quad \underline{\underline{AP}}$$

$$(n) \quad \frac{3}{\sqrt{2}} + 2\sqrt{2} + \frac{5}{\sqrt{2}} + \dots \text{ to 15 terms}$$

S1

$$\therefore S_n = \frac{n}{2} \{2a + (n-1)d\}$$

$$\text{here, } a = \frac{3}{\sqrt{2}}, n = 15,$$

$$d = \frac{2\sqrt{2} - \frac{3}{\sqrt{2}}}{1} = \frac{5 - 2\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$\therefore S_n = \frac{15}{2} \left\{ 2 \left(\frac{3}{\sqrt{2}} \right) + (15-1) \frac{\sqrt{2}}{2} \right\}$$

(32) Find number of terms of following -

$$(-1) + (-5) + (-9) + \dots, \text{ if sum} = 65$$

S1

$$\therefore S_n = \frac{n}{2} \{2a + (n-1)d\}$$

$$\text{here, } S_n = 65, a = -1$$

$$d = -5 - (-1) = -5 - (-1) = -4$$

$$\therefore 65 = \frac{n}{2} \{2(-1) + (n-1)(-4)\}$$

$$\Rightarrow 130 = n(-14 + 2n - 2)$$

$$\Rightarrow 130 = n(-16 + 2n)$$

$$\Rightarrow 130 = -16n + 2n^2$$

$$\Rightarrow 0 = 2n^2 - 16n - 130$$

$$\Rightarrow 2n^2 - 16n + 10n - 130 = 0$$

$$\Rightarrow 2^n(n-13) + 10(n-13) = 0$$

$$\Rightarrow (n-13)(n+10) = 0$$

Either,

$$n-13 = 0 \quad \text{or} \quad n+10 = 0$$

$$n = 13$$

$$n = -10$$

Thus,

$$\boxed{n = 13}$$

(32)

(ii) $-7 + (-4) + (-1) + \dots$; if sum is 114

Sol

$$\therefore S_n = \frac{n}{2} [2a + (n-1)d]$$

$$\text{here, } S_n = 114, a = -7,$$

$$d = -4 - (-7) = -1 - (-4) = 3$$

$$\therefore 114 = \frac{n}{2} [2(-7) + (n-1)3]$$

$$\Rightarrow 228 = n(-14 + 3n - 3)$$

$$\Rightarrow 228 = -17n + 3n^2$$

$$\Rightarrow 3n^2 - 17n - 228 = 0$$

$$\Rightarrow 3n^2 - 36n + 19n - 228 = 0$$

$$\Rightarrow 3n(n-12) + 19(n-12) = 0$$

$$\Rightarrow (n-12)(3n+19) = 0$$

Either,

$$n-12=0 \text{ or } 3n+19=0$$

$$n=12$$

$$n = -\frac{19}{3}$$

Thus, $n=12$ Ans(iii) $-9 + (-6) + (-3) + \dots$; if sum = 3225

Sol

$$\text{Here, } a = -9, S_n = 3225$$

$$d = -6 - (-9) = -3 - (-6) = 3$$

$$\therefore S_n = \frac{n}{2} [2a + (n-1)d]$$

$$\Rightarrow 3225 = \frac{n}{2} [2(-9) + (n-1)3]$$

$$\Rightarrow 6450 = n(-18 + 3n - 3)$$

$$\Rightarrow 6450 = -21n + 3n^2$$

$$\Rightarrow 3n^2 - 21n - 6450 = 0$$

$$\Rightarrow 3n^2 - 150n + 129n - 6450 = 0$$

$$\Rightarrow 3n(n-50) + 129(n-50) = 0$$

$$\Rightarrow (n-50)(3n+129) = 0$$

Either,

$$n-50=0 \text{ or } 3n+129=0$$

$$n=50$$

$$n = -\frac{129}{3}$$

Thus, $n=50$ Ans(iv) $-9 - 6 - 3 + 0 + \dots$; if sum = 66

Sol

$$\text{Here, } a = -9, S_n = 66,$$

$$d = -6 - (-9) = -3 - (-6) = 3$$

$$\therefore 66 = \frac{n}{2} [2(-9) + (n-1)3]$$

$$\Rightarrow 132 = n(-18 + 3n - 3)$$

$$\Rightarrow 3n^2 - 21n - 132 = 0$$

$$\Rightarrow 3n^2 - 33n + 22n - 132 = 0$$

$$\Rightarrow 3n(n-11) + 22(n-11) = 0$$

$$\Rightarrow (n-11)(3n+22) = 0$$

Either,

$$n-11=0$$

$$3n+22=0$$

$$n=11$$

$$n = -\frac{22}{3}$$

Thus, $n=11$ Ans

③ Sum the series:

① $3+5-7+9+11-13+15+17-19+\dots$ to $3n$ terms.

We can also write it as:

$$= 3+9+15+\dots \text{ to } n \text{ term}$$

$$= 5+11+17+\dots \text{ to } n \text{ term}$$

$$= -7-13-19+\dots \text{ to } n \text{ term}$$

$$\therefore S_n = S_n' + S_n'' + S_n''' \quad \text{--- (1)}$$

For S_n' :

$$3+9+15+\dots \text{ to } n \text{ term}$$

$$\text{Here, } a=3, n=n, \&$$

$$d=9-3=15-9=6$$

$$\therefore S_n' = n/2 \{2(3) + (n-1)6\}$$

$$\Rightarrow S_n' = n/2 (8+6n-6)$$

$$\Rightarrow S_n' = 3n^2$$

For S_n'' :

$$5+11+17+\dots \text{ to } n \text{ term}$$

$$\text{Here, } a=5, n=n, \&$$

$$d=11-5=17-11=6$$

$$\therefore S_n'' = n/2 \{2(5) + (n-1)6\}$$

$$\Rightarrow S_n'' = n/2 (10+6n-6)$$

$$\Rightarrow S_n'' = n/2 (4+6n)$$

$$\Rightarrow S_n'' = 2n+3n^2$$

For S_n''' :

$$-7-13-19+\dots \text{ to } n \text{ term}$$

$$\text{Here, } a=-7, n=n, \&$$

$$d=-13-(-7)=-19-(-13)=-6$$

$$\therefore S_n''' = n/2 \{2(-7) + (n-1)(-6)\}$$

$$\Rightarrow S_n''' = n/2 (-14-6n+6)$$

$$\Rightarrow S_n''' = n/2 (-8-6n)$$

$$\Rightarrow S_n''' = -4n-3n^2$$

Put values of S_n', S_n''

and S_n''' in (1)

$$\text{①} \Rightarrow S_n = 3n^2 + (2n+3n^2) + (-4n-3n^2)$$

$$\Rightarrow S_n = 3n^2 - 2n$$

$$\Rightarrow S_n = n(3n-2)$$

(34)

(iii) $1+4+7+10+13+16+19+22+25+\dots$ to $2n$ terms.

S_1

We can also write it as
 $= 1+10+19+\dots$ to n term
 $= 4+13+22+\dots$ to n term
 $= -7-16-25+\dots$ to n term

$$\therefore S_n = S_n^I + S_n^{II} + S_n^{III} \quad \text{--- (1)}$$

For S_n^I :

$1+10+19+\dots$ to n term

Here, $a=1$, $n=n$ &

$$d = 10-1 = 19-10 = 9$$

$$\therefore S_n^I = \frac{n}{2} \{2(1) + (n-1)9\}$$

$$\Rightarrow S_n^I = \frac{n}{2} (2+9n-9)$$

$$\Rightarrow S_n^I = \frac{9n^2 - 7n}{2}$$

For S_n^{II} :

$4+13+22+\dots$ to n term

Here, $a=4$, $n=n$ and

$$d = 13-4 = 22-13 = 9$$

$$\therefore S_n^{II} = \frac{n}{2} \{2(4) + (n-1)9\}$$

$$\Rightarrow S_n^{II} = \frac{n}{2} (8+9n-9)$$

$$\Rightarrow S_n^{II} = \frac{n}{2} (1+9n)$$

$$\Rightarrow S_n^{II} = \frac{-n+9n^2}{2}$$

For S_n^{III} :

$-7-16-25+\dots$ to n term

Here, $a=-7$, $n=n$ and

$$d = -16-(-7) = -25-(-16) = -9$$

$$\therefore S_n^{III} = \frac{n}{2} \{2(-7) + (n-1)(-9)\}$$

$$\Rightarrow S_n^{III} = \frac{n}{2} (-14-9n+9)$$

$$\Rightarrow S_n^{III} = \frac{n}{2} (-5-9n)$$

$$\Rightarrow S_n^{III} = \frac{-5n-9n^2}{2}$$

$$\therefore \text{①} \Rightarrow \frac{9n^2-7n}{2} + \frac{(-n+9n^2)}{2} + \frac{(-5n-9n^2)}{2}$$

$$\Rightarrow S_n = \frac{9n^2-7n-n+9n^2-5n-9n^2}{2}$$

$$\Rightarrow S_n = \frac{9n^2-13n}{2}$$

$$\Rightarrow S_n = \frac{n(9n-13)}{2} \quad \frac{dp}{dx}$$

(iii) $9+12+(-15)+18+21-24+27+30-33+\dots$ to $3n$ terms.

Sol

We can also write it as:

$$= 9+18+27+\dots \text{ to } n \text{ term}$$

$$= 12+21+30+\dots \text{ to } n \text{ term}$$

$$= -15+(-24)+(-33)+\dots \text{ to } n \text{ term}$$

$$\therefore S_n = S_n' + S_n'' + S_n'''$$

For S_n' :

$$= 9+18+27+\dots \text{ to } n \text{ term}$$

$$\text{Here, } a = 9, n = n \text{ \& } d = 18-9 = 27-18 = 9$$

$$\therefore S_n' = \frac{n}{2} \{2(9) + (n-1)9\}$$

$$\Rightarrow S_n' = \frac{n}{2} (18 + 9n - 9)$$

$$\Rightarrow S_n' = \frac{n}{2} (9 + 9n)$$

$$\Rightarrow S_n' = \frac{9n + 9n^2}{2}$$

For S_n'' :

$$= 12+21+30+\dots \text{ to } n \text{ term}$$

$$\text{Here, } a = 12, n = n \text{ and } d = 21-12 = 30-21 = 9$$

$$\therefore S_n'' = \frac{n}{2} \{2(12) + (n-1)9\}$$

$$\Rightarrow S_n'' = \frac{n}{2} (24 + 9n - 9)$$

$$\Rightarrow S_n'' = \frac{n}{2} (15 + 9n)$$

$$\Rightarrow S_n'' = \frac{15n + 9n^2}{2}$$

$$\Rightarrow S_n'' = \frac{15n + 9n^2}{2}$$

For S_n''' :

$$= -15-24-33+\dots \text{ to } n \text{ term}$$

$$\text{Here, } a = -15, n = n \text{ \& } d = -24 - (-15) = -33 - (-24) = -9$$

$$\therefore S_n''' = \frac{n}{2} \{2(-15) + (n-1)(-9)\}$$

$$\Rightarrow S_n''' = \frac{n}{2} (-30 - 9n + 9)$$

$$\Rightarrow S_n''' = \frac{n}{2} (-21 - 9n)$$

$$\Rightarrow S_n''' = \frac{-21n - 9n^2}{2}$$

$$\Rightarrow S_n''' = \frac{-21n - 9n^2}{2}$$

Put in (1).

$$\therefore \textcircled{1} \Rightarrow \frac{9n + 9n^2}{2} + \frac{15n + 9n^2}{2} + \frac{(-21n - 9n^2)}{2}$$

$$\Rightarrow \frac{9n + 9n^2 + 15n - 21n}{2}$$

$$\Rightarrow \frac{3n + 9n^2}{2}$$

$$\Rightarrow S_n = \frac{3n(1 + 3n)}{2}$$

(36)

④ If $S_n = n(2n-1)$, then find the series.

Sol

Given: $S_n = n(2n-1)$

For $n=1$ we get,

$$\Rightarrow S_1 = 1[2(1)-1] = 1$$

For $n=2$ we get,

$$\Rightarrow S_2 = 2[2(2)-1] = 6$$

For $n=3$ we get

$$\Rightarrow S_3 = 3[2(3)-1] = 15$$

For $n=4$ we get,

$$\Rightarrow S_4 = 4[2(4)-1] = 28$$

For first term.

$$\Rightarrow T_1 = S_1 = 1$$

For 2nd term.

$$\Rightarrow T_2 = S_2 - S_1 = 6 - 1 = 5$$

For third term.

$$\Rightarrow T_3 = S_3 - S_2 = 15 - 6 = 9$$

For fourth term.

$$\Rightarrow T_4 = S_4 - S_3 = 28 - 15 = 13$$

Thus $1 + 5 + 9 + 13 + \dots$ is the required series. Ans

⑤ Find sum of first 100 natural numbers which are neither exactly divisible by 3 nor by 7.

Sol

For first 100 natural numbers.

$$= 1 + 2 + 3 + \dots + 100$$

$$\therefore a_n = a + (n-1)d$$

$$\therefore 100 = 1 + (n-1)1$$

$$\Rightarrow 100 = 1 + n - 1$$

$$\Rightarrow n = 100$$

Now,

$$S_n' = n/2 \{a + l\}$$

$$S_n' = 100/2 \{1 + 100\}$$

$$S_n' = 5 \times (101)$$

$$S_n' = 5050$$

(37)

For numbers divisible by 3 out of first 100 natural numbers.

$$= 3 + 6 + 9 + \dots + 99$$

$$\therefore a_n = a + (n-1)d$$

$$\therefore 99 = 3 + (n-1)3$$

$$\Rightarrow 99 = 3 + 3n - 3$$

$$\Rightarrow n = 99/3$$

$$\Rightarrow n = 33$$

Now,

$$\Rightarrow S_n = n/2 (a + l)$$

$$\Rightarrow S_n = 33/2 (3 + 99)$$

$$\Rightarrow S_n = 33/2 (102)$$

$$\Rightarrow S_n = 33(56)$$

$$\Rightarrow S_n = 1848$$

For numbers divisible by 7 out of first 100 natural numbers.

$$= 7 + 14 + 21 + \dots + 98$$

$$\therefore a_n = a + (n-1)d$$

$$\therefore 98 = 7 + (n-1)7$$

$$\Rightarrow 98 = 7 + 7n - 7$$

$$\Rightarrow n = 98/7$$

$$\Rightarrow n = 14$$

Now,

$$\Rightarrow S_n = n/2 (a + l)$$

$$\Rightarrow S_n = 14/2 (7 + 98)$$

$$\Rightarrow S_n = 7(105)$$

$$\Rightarrow S_n = 735$$

For numbers divisible by both 3 and 7

$$= 21 + 42 + 63 + \dots + 84$$

$$\therefore a_n = a + (n-1)d$$

$$\therefore 84 = 21 + (n-1)21$$

$$\Rightarrow 84 = 21 + 21n - 21$$

$$\Rightarrow n = 84/21$$

$$\Rightarrow n = 4$$

Now,

$$\Rightarrow S_n = n/2 (a + l)$$

$$\Rightarrow S_n = 4/2 (21 + 84)$$

$$\Rightarrow S_n = 2(105)$$

$$\Rightarrow S_n = 210$$

$$S_n = S_n' - S_n'' - S_n''' + S_n^{IV}$$

$$\Rightarrow S_n = 5050 - 1683 - 735 + 210$$

$$\Rightarrow S_n = 2842$$

Q.P.
=.

Q) If S_2, S_3, S_5 are the sum of $2n, 3n$ and $5n$ terms of an A.P. - Show that $S_5 = 5(S_3 - S_2)$

S.I.

$$\text{let, } S_n = a + (a+d) + (a+2d) + (a+3d) + \dots + a + (n-1)d$$

$$\therefore S_n = \frac{n}{2} [2a + (n-1)d]$$

Therefore,

$$\Rightarrow S_2 = \frac{2n}{2} [2a + (2n-1)d]$$

$$\Rightarrow S_3 = \frac{3n}{2} [2a + (3n-1)d]$$

$$\Rightarrow S_5 = \frac{5n}{2} [2a + (5n-1)d]$$

Show that : $S_5 = 5(S_3 - S_2)$

$$\begin{aligned} \Rightarrow \frac{5n}{2} [2a + (5n-1)d] &= 5 \left[\frac{3n}{2} [2a + (3n-1)d] - \frac{2n}{2} [2a + (2n-1)d] \right] \\ &= \frac{5n}{2} [3(2a + 3nd - d) - 2(2a + 2nd - d)] \\ &= \frac{5n}{2} (6a + 9nd - 3d - 4a - 4nd + 2d) \\ &= \frac{5n}{2} (2a + 5nd - d) \end{aligned}$$

$$\Rightarrow \frac{5n}{2} [2a + (5n-1)d] = \frac{5n}{2} [2a + (5n-1)d] \quad \underline{\text{proved!}}$$

Q) The sum of n terms of two arithmetic series are in the ratio of $3n+2n:n+1$. Find the ratio of their 2^{nd} term.

S.I.
=

(39)

let S_n' and S_n'' are the two A.P i.e. S

$$S_n' = a_1 + (a_1 + d_1) + (a_1 + 2d_1) + (a_1 + 3d_1) + \dots + a_1 + (n-1)d_1$$

$$\therefore S_n = \frac{n}{2} [2a_1 + (n-1)d_1]$$

$$\therefore S_n' = \frac{n}{2} [2a_1 + (n-1)d_1]$$

For eight term,

$$\therefore T_n = a + (n-1)d_1$$

$$\therefore T_8' = a_1 + 7d_1$$

$$S_n'' = a_2 + (a_2 + d_2) + (a_2 + 2d_2) + (a_2 + 3d_2) + \dots + a_2 + (n-1)d_2$$

$$\therefore S_n = \frac{n}{2} [2a_2 + (n-1)d_2]$$

$$\therefore S_n'' = \frac{n}{2} [2a_2 + (n-1)d_2]$$

For eight term,

$$\therefore T_n = a + (n-1)d$$

$$\therefore T_8'' = a_2 + 7d_2$$

$\therefore$ It is given that:

$$S_n' : S_n'' = 3n+2 : n+1$$

$$\therefore \frac{\frac{n}{2} [2a_1 + (n-1)d_1]}{\frac{n}{2} [2a_2 + (n-1)d_2]} = \frac{3n+2}{n+1}$$

$$\Rightarrow \frac{2[a_1 + (\frac{n-1}{2})d_1]}{2[a_2 + (\frac{n-1}{2})d_2]} = \frac{3n+2}{n+1} \quad \text{--- (1)}$$

For eight term $\frac{n-1}{2}$ should equal to 7.

$$\therefore \frac{n-1}{2} = 7$$

$$\Rightarrow n = 14 + 1$$

$$\Rightarrow n = 15$$

Put $n = 15$ in (1)

$$\textcircled{1} \Rightarrow \frac{a_1 + \left(\frac{15-1}{2}\right)d_1}{a_2 + \left(\frac{15-1}{2}\right)d_2} = \frac{3(15)+2}{75+1}$$

$$\Rightarrow \frac{a_1 + 7d_1}{a_2 + 7d_2} = \frac{45+2}{16} \quad \text{--- } \textcircled{2}$$

$$\therefore a_1 + 7d_1 = T_8' \quad \text{and} \quad a_2 + 7d_2 = T_8''$$

$$\therefore \frac{T_8'}{T_8''} = \frac{47}{16}$$

$$\text{or } T_8' : T_8'' = 47 : 16$$

AP

- ⑧ The sum of three numbers in A.P. are 27 and their product is 405. Find the numbers.

Sol

Let, $(a-d) + a + (a+d)$ are the three numbers in A.P.

A/c to given condition:

$$a-d + a + a+d = 27 \quad \text{--- } \textcircled{1}$$

$$(a-d)(a)(a+d) = 405 \quad \text{--- } \textcircled{2}$$

Take ①

$$\textcircled{1} \Rightarrow a-d+a+a-d=27$$

$$\Rightarrow 3a=27$$

$$\Rightarrow a=9$$

Put in ②

$$\textcircled{2} \Rightarrow (a-d)(a)(a+d)=405$$

$$\Rightarrow (a-d)(a+d) = 405/a$$

$$\Rightarrow (9)^2 - (d)^2 = 45$$

$$\Rightarrow 81 - d^2 = 45$$

$$\Rightarrow 36 = d^2$$

$$\Rightarrow d = \pm 6$$

$$\text{When } a=9 \text{ and } d=-6$$

$$1^{\text{st}} \text{ no} = a-d = 9-(-6)=15$$

$$2^{\text{nd}} \text{ no} = a = 9$$

$$3^{\text{rd}} \text{ no.} = a+d = 9+(-6)=3$$

Thus, When $a=9$ & $d=-6$

$$\text{First number} = a-d = 9-6=3$$

$$\text{Second number} = a = 9$$

$$\text{Third number} = a+d = 9+6=15$$

Required 3 no. are either

3, 9, 15 or 15, 9, 3

④ Find five numbers in A.P. whose sum is 25 and the sum of whose squares is 135

Sol.

Let, $a-2d, a-d, a, a+d, a+2d$ are five number.

A/c to given condition:

$$\Rightarrow (a-2d) + (a-d) + (a) + (a+d) + (a+2d) = 25 \quad \text{--- ①}$$

$$\Rightarrow (a-2d)^2 + (a-d)^2 + (a)^2 + (a+d)^2 + (a+2d)^2 = 135 \quad \text{--- ②}$$

Take ①

42

$$\textcircled{1} \Rightarrow a - 2d + a - d + a + a + d + a + 2d = 25$$

$$\Rightarrow 5a = 25$$

$$\Rightarrow a = 5$$

Put $a = 5$ in ②

$$\textcircled{2} \Rightarrow (5-2d)^2 + (5-d)^2 + (5)^2 + (5+d)^2 + (5+2d)^2 = 135$$

$$\Rightarrow 25 - 20d + 4d^2 + 25 - 10d + d^2 + 25 + 25 + 10d + d^2 + 25 + 20d + 4d^2 = 135$$

$$\Rightarrow 125 + 10d^2 = 135$$

$$\Rightarrow 10d^2 = 10$$

$$\Rightarrow d = \sqrt{1}$$

$$\Rightarrow d = \pm 1$$

If $a = 5$ and $d = 1$

$$1^{\text{st}} \text{ no.} = a - 2d = 5 - 2(1) = 3$$

$$2^{\text{nd}} \text{ no.} = a - d = 5 - 1 = 4$$

$$3^{\text{rd}} \text{ no.} = a = 5$$

$$4^{\text{th}} \text{ no.} = a + d = 5 + 1 = 6$$

$$5^{\text{th}} \text{ no.} = a + 2d = 5 + 2(1) = 7$$

If $a = 5$ and $d = -1$

$$1^{\text{st}} \text{ no.} = a - 2d = 5 - 2(-1) = 7$$

$$2^{\text{nd}} \text{ no.} = a - d = 5 - (-1) = 6$$

$$3^{\text{rd}} \text{ no.} = a = 5$$

$$4^{\text{th}} \text{ no.} = a + d = 5 + (-1) = 4$$

$$5^{\text{th}} \text{ no.} = a + 2d = 5 + 2(-1) = 3$$

Thus the required 5 no. are
either 3, 4, 5, 6, 7 or 7, 6, 5, 4, 3

Ans

⑩ The sum of Rs. 42000 is distributed among 5 Persons so that each person after the first receive Rs. 20 less than the preceeding person. How much does each person receive?

(43)

Given: total sum (S_n) = 42000, no. of persons (n) = 5,
common diff (d) = -80.

$\therefore$ each person after the 1st person receives Rs. 80 less than proceeding person - It shows that the difference of amount among each person is common. So it is an A.P.

$$\therefore S_n = a + a + d + a + 2d + \dots + a + (n-1)d$$

$$\Rightarrow S_n = \frac{n}{2} \{2a + (n-1)d\}$$

$$\Rightarrow 42000 = \frac{5}{2} \{2a + (5-1)(-80)\}$$

$$\Rightarrow 84000 = 5(2a - 320)$$

$$\Rightarrow 16800 = 2a - 320$$

$$\Rightarrow 2a = 16800 + 320$$

$$\Rightarrow a = \frac{17120}{2}$$

$$\Rightarrow a = 8560$$

Here,

$$A_1 = a = 8560$$

$$A_2 = a + d = 8560 - 80 = 8480$$

$$A_3 = a + 2d = 8560 + 2(-80) = 8400$$

$$A_4 = a + 3d = 8560 + 3(-80) = 8320$$

$$A_5 = a + 4d = 8560 + 4(-80) = 8240$$

Thus,

Amount received by 1st Person (A_1) = 8560

" " " 2nd Person (A_2) = 8480

" " " 3rd Person (A_3) = 8400

" " " 4th Person (A_4) = 8320

" " " 5th Person (A_5) = 8240

(44)

Q. A well digging company charges Rs. 1250 for the first meter, Rs. 1500 for the second and Rs. 1750 for the third meter and so on. What is the depth of well that cost Rs. 50,000?

$$S_n =$$

$$\text{Charges for first meter} = a_1 = 1250$$

$$\text{Charges for Second meter} = a_2 = 1500$$

$$\text{" " third meter} = a_3 = 1750$$

$$\text{Total cost } (S_n) = 50,000.$$

$$d = a_2 - a_1 = 1500 - 1250 = 250$$

$$\text{or } d = a_3 - a_2 = 1750 - 1500 = 250$$

$\therefore$ difference is common

$\therefore$ It is an A.P. i.e.

$$S_n = a + a + d + a + 2d + \dots + a + (n-1)d$$

$$S_n = \frac{n}{2} \{2a + (n-1)d\}$$

$$50000 = \frac{n}{2} \{2(1250) + (n-1)250\}$$

$$100000 = n(2500 + 250n - 250)$$

$$100000 = n(2250 + 250n)$$

$$100000 = 250n(a + n)$$

$$400 = an^2 + n^2$$

$$0 = n^2 + an + 400$$

$$0 = n^2 + 25n - 16n + 400$$

$$0 = n(n+25) - 16(n+25)$$

$$\Rightarrow (n+25)(n-16) = 0$$

Either,

$$n+25=0 \text{ or } n-16=0$$

$$n = -25$$

$$\boxed{n = 16}$$

$\therefore$ depth can't be "ve"

Thus,

$$\boxed{n = 16}$$

Ans

(45)

A man borrows Rs 25,000 and agrees to repay with a total profit of 10,000 in 10 installments, each installment being less than the preceding by Rs. 200. What should be his first installment?

Sol

Since each installment is less than the preceding by Rs. 200 i.e. $d = -200$

$$\text{Total payment} = \{S_n\} = 25000 + 10000 = 35000$$

$$\text{no. of installments } (n) = 10.$$

$$a = ?$$

$$\therefore S_n = n/2 \{2a + (n-1)d\}$$

$$\therefore 35000 = 10/2 \{2a + (10-1) \cdot -200\}$$

$$\Rightarrow 35000 = 5(2a - 1800)$$

$$\Rightarrow \frac{35000}{5} = 2a - 1800$$

$$\Rightarrow 2a = 7000 + 1800$$

$$\Rightarrow a = \frac{8800}{2}$$

$$\Rightarrow a = 4400$$

Thus first installment should be 4400

or
=